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A Multi-Institutional Study of High School Mathematics Curricula and College Mathematics Achievement and Course Taking

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This study examined the relationship between high school mathematics curricula and student achievement and course-taking patterns over 4 years of college course taking for a sample of over 10,000 students from 32 postsecondary 4-year institutions. Three types of curricula were studied: National Science Foundation (NSF) funded curricula, the University of Chicago School Mathematics Project curriculum, and commercially developed curricula. The major result was that high school mathematics curricula were unrelated to college mathematics achievement or students' course-taking patterns for students who began college with precalculus (college algebra) or a more difficult course. However, students of the NSF-funded curricula were statistically more likely to begin their college mathematics at the developmental level. Implications of these results for research and practice are discussed.

Keywords: Achievement; College mathematics; Integrated curricula; Large-scale studies; Longitudinal studies; Mathematics education

Surprisingly little is known about the relationship between high school mathematics curricula and student preparation for college mathematics, prompting calls for studies that clarify this relationship (Augustine, 2007; Mathematical Sciences Education Board [MSEB], 2004). Such studies are important for three related reasons. First, it is important to establish an empirical research base characterizing the likelihood that various curricula differentially prepare students for college mathematics (MSEB, 2004; Planty et al., 2008), particularly in light of evidence that many students are not prepared to succeed at this level (Planty et al., 2008; Snyder, Dillow, & Hoffman, 2009).[1] Second, it is important to examine the ability of high school mathematics curricula to prepare an increasingly diverse group of students for college mathematics in postsecondary institutions that vary in enrollment, selectivity, and educational and financial profiles. Third, there is significant interest in identifying predictors of developmental mathematics course enrollment. Developmental mathematics courses address content that is ideally covered in high school and do not usually count toward college degree completion (Adelman, 2004).

[1] We define *college* as a 4-year postsecondary institution offering a bachelor's degree.

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Approximately 25% of college freshmen complete at least one developmental mathematics course (Parsad & Lewis, 2003), and many states are considering postsecondary policies related to mathematics remediation (Bettinger & Long, 2009). Current literature is largely limited to using student outcomes in high school, such as GPA and percentile rank, as predictors of developmental mathematics course taking in college (Attewell, Lavin, Domina, & Levey, 2006; Bettinger & Long, 2009); however, none have studied the role of high school mathematics curriculum on developmental course taking.

The purpose of this study is to deepen our understanding of the relationship between high school mathematics curricula and college mathematics achievement and course taking for a diverse group of students and postsecondary institutions. We begin by briefly describing the high school mathematics curricula we studied, and then we summarize available literature on the relationship between these curricula and student achievement and course taking in college mathematics, including developmental courses. After posing research questions reflecting the study's purpose, we describe our methodology and findings. We conclude by discussing the implications of our results and suggestions for additional research.

High School Mathematics Curricula

The development of almost all high school mathematics curricula currently used in the United States can be traced to one of three sources: the National Science Foundation, for-profit companies (usually publishers), or a combination of the two. High school mathematics curricula linked to these sources share several features, but they also differ in important ways in their approach to mathematics content, instructional practices, student assessment, and in the populations of students the curricula were designed for (e.g., students who will take advanced mathematics courses and those who will not). These differences have led to concerns about the ability of different curricula to prepare students for college mathematics, especially calculus (Schoenfeld, 2004).

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National Science Foundation Funded Curricula

National Science Foundation (NSF) funded curricula represent those funded through a proposal solicitation in the early 1990s (NSF RFP 91-100). The expectation was that the development of new curricula would be guided by the *Curriculum and Evaluation Standards for School Mathematics* (National Council of Teachers of Mathematics [NCTM], 1989). In many ways this was a theory-based, best practices document in its incorporation of findings from cognitively oriented research on human learning and instructional practices (Bruner, 1964; Shulman & Sparks, 1992). The Standards also focused on engaging students in a learning environment whose features include group work and problem solving motivated by real-world problems and data (Schoenfeld, 2004). The current study included three NSF-funded first edition high school curricula: (a) Contemporary Mathematics in Context (CMIC or Core-Plus), (b) Interactive Mathematics Program (IMP), and (c) Mathematics: Modeling Our World (MMOW). These curricula are similar in their focus on algebra, geometry, probability, statistics, and topics in discrete mathematics every school year in a way that emphasizes problem solving and deemphasizes algorithmic manipulation. These curricula also share a heavy reliance on contextually oriented mathematical activity with real-world problems selected because of their high potential for the development of important mathematical topics.

Commercially Developed Curricula

In the late 20th and early 21st centuries, commercially developed curriculum was the most widely used curriculum type, reaching approximately 75% of the nation's students (Education Market Research, 2006). Commercially developed (commercial) curricula are sometimes referred to as *traditional curricula* because of their focus on longstanding conventions about which mathematics topics students should learn and how this material should be learned. These curricula partitioned the high school mathematics program into algebra (Year 1), geometry (Year 2), and a second course in algebra (Year 3). They tended to be teacher centered and depended more heavily on algorithmic development and repetition (Schoenfeld, 2002; Schoen, Fey, Hirsch, & Coxford, 1999). For college-intending students, these curricula typically included a senior year precalculus (college algebra) course or, in many cases, a first year Advanced Placement Calculus course. Examples of publishers whose programs are included in this category are Addison Wesley, Saxon, Glencoe/McGraw Hill, and Scott Foresman. These curricula represent a distinct alternative to NSF-funded curricula approaches.

University of Chicago School Mathematics Project Curriculum

Another category of curriculum is represented by the University of Chicago School Mathematics Project (Usiskin, 1986). This program combines features of both NSF-funded and commercial curricula (Hirschhorn, 1993; Schoenfeld, 2004; Thompson & Senk, 2001). In our estimation, UCSMP leans more heavily in the direction of the NSF approaches.

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The UCSMP curriculum has received funding from private sources (e.g., Amoco Corporation) as well as NSF support (Education Market Research, 2006). It is best known through the algebra, geometry, algebra II, functions, statistics, trigonometry, precalculus, and discrete mathematics texts that emphasize reading, problem solving, typical or everyday applications of mathematics, and the use of technology, including spreadsheets, calculators, and computers. The ongoing development of the UCSMP curriculum has been informed by a theoretical model that incorporates research findings on student learning.

The Relationship Between High School Mathematics Curricula and Student Preparation for College Mathematics

Research examining the relationship between high school mathematics curricula and preparation for college mathematics has typically focused on the NSF-funded curricula, largely as a response to criticism of these curricula as detailed in Schoenfeld (2004). Much of the criticism claims that NSF-funded curricula do not adequately prepare college-bound students to succeed in calculus and other advanced college mathematics courses (Klein, 2000; Wu, 1997). Yet only a few studies have examined the impact of high school mathematics curriculum on college mathematics achievement and course taking, both nondevelopmental and developmental.

Schoen and Hirsch (2003) examined the college Calculus I grades of students from two similar high schools, one of which offered an NSF-funded curriculum (Core-Plus) and the other a commercial curriculum. Students who completed Core-Plus had higher grades in the college Calculus I course and in subsequent mathematics courses when compared with students who completed a commercial program. On the other hand, Harwell et al. (2009) used a sample of more than 3,000 students and reported no difference in average grades earned in the first college mathematics class of students who completed an NSF-funded or commercial curriculum when variables capturing a student's background such as ethnicity and prior ability (e.g., high school mathematics GPA) were taken into account.

Evidence of the impact of curriculum on college mathematics course taking suggests that students who completed an NSF-funded curriculum tended to initially enroll in less difficult (nondevelopmental) college mathematics courses than students who completed a commercial curriculum (Harwell et al., 2009). Post et al. (2010) probed this finding by examining the relationship between curriculum and nondevelopmental mathematics course taking for each of three strata created using ACT mathematics scores. The results showed that students in the lowest stratum who completed an NSF-funded curriculum tended to enroll in less difficult college mathematics courses (e.g., precalculus versus calculus) than those who completed a commercial curriculum, but for students in higher strata no relationship emerged.

Additional evidence reflecting a student's preparation for college mathematics comes from examining the likelihood of completing a developmental mathematics course. A significant amount of literature has documented the role that high school

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mathematics achievement plays in developmental mathematics course taking in college (e.g., Adelman, 2004; Bettinger & Long, 2009), but only two studies are currently available that address the relationship between specific high school mathematics curricula and developmental course taking in college mathematics.

Harwell et al. (2009) used data from a single postsecondary institution and found that students who completed an NSF-funded curriculum were statistically more likely to initially enroll in a developmental mathematics course in college compared to students who completed a commercial curriculum; among students who began with a nondevelopmental (higher difficulty) course, no differences among the curriculum cohorts emerged. Also, when students' ACT mathematics scores were partitioned into thirds, there were no differences in course-taking patterns among the curriculum cohorts for students in the middle and high ACT levels (Post et al., 2010).

A related area of research has focused on the relationship between student ethnicity and mathematics achievement and course taking. It is well known that a mathematics achievement gap exists between African American and white students in high school (Braun, Wang, Jenkins, & Weinbaum, 2006; Riegle-Crumb & Grodsky, 2010; Snyder et al., 2009) and college (Adelman, 2006; Spenner, Buchmann, & Landerman, 2004; Vars & Bowen, 1998). Studying the impact of high school mathematics curricula is important because of arguments that the achievement gap can in part be attributed to instructional practices within schools that can be modified (Gamoran & Berends, 1987; Snyder, 2007). Furthermore, one common goal of NSF-funded, commercial, and UCSMP curricula is to make mathematics accessible to all students (Thompson & Senk, 2001), which would be expected to affect the achievement gap.

In summary, a few studies have examined the impact of high school mathematics curricula on students' preparation for college mathematics and have provided insight into this important relationship. These studies have relied on sampling students from a single postsecondary institution and—with the exception of the Schoen and Hirsch (2003) study that used random assignment of curriculum to high schools but with a small sample—employed various statistical methods to try to strengthen causal inferences. Moreover, none of these studies examined students' achievement and the difficulty level of their mathematics courses over their 4-year college career, including their performance in their last college mathematics course. The latter is an important benchmark in the sense that where (and how) students finish their college mathematics is arguably more important than where they began.

To better understand the relationship between high school mathematics curricula and college mathematics achievement and course taking for a diverse group of students and postsecondary institutions, we posed the following research questions:

1. What is the nature and magnitude of the relationship between the high school mathematics curriculum completed and a student's (a) grade in college Calculus I, (b) grade in last college mathematics course, and (c) mathematics course grades across eight college semesters?

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2. What is the nature and magnitude of the relationship between the high school mathematics curriculum students complete and the (a) likelihood that the difficulty level of their last college mathematics course equaled or exceeded that of Calculus I, (b) pattern of difficulty levels of their college mathematics course(s) when difficulty equaled or exceeded college Calculus I across eight semesters, and (c) likelihood that they completed a developmental mathematics course in college?
3. Do the relationships in the research questions above differ across ethnic groups and postsecondary institutions varying in enrollment, selectivity, and educational and financial profiles?

We included a student's college Calculus I class grade and the likelihood that the difficulty level of a student's last college mathematics course equaled or exceeded that of Calculus I as outcomes to provide an empirical test of the ability of the three high school mathematics programs to prepare students for advanced college mathematics courses. The likelihood that a student completed a developmental mathematics course was included because of the high percentage of students in the United States who completed at least one developmental mathematics course, 16% in public colleges and 8% in private colleges (Parsad & Lewis, 2003). Also, a difference among curricula would imply a difference in student preparation for college mathematics. Question 3 was included because of literature documenting the impact of student ethnicity on college mathematics performance (Adelman, 2006; Spenner et al., 2004). We note that SES was not included because good measures of a high school student's SES (e.g., parental occupation or education) were not available.

Method

Research Design

A retrospective cohort (nonequivalent, posttest only, quasi-experimental) cluster design was used in which archival data, including college mathematics achievement and course-taking data covering eight semesters, were obtained for students who enrolled at one of 32 postsecondary institutions (clusters) in Fall 2002 or Fall 2003. Students who completed the same high school mathematics curriculum were treated as members of the same curriculum cohort. Following the What Works Clearinghouse (2008) standards, preexisting differences among cohorts were taken into account using control variables in the data analyses. This helps to ensure, but does not guarantee, credible inferences. We also acknowledge that our use of a quasi-experimental design implies that our study does not meet all of the What Works Clearinghouse standards for estimating an intervention (curriculum) effect, but we believe that our design and analyses produce credible answers to our research questions.

Sampling

The target population consisted of colleges in the United States, whereas the sampled population consisted of colleges in the upper Midwest that differed in

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several ways. The 46 colleges we invited to participate varied in enrollment (full-time and part-time students), selectivity (ACT score of the freshman class corresponding to the 25th percentile), and financial profile (publicly or privately supported), based on a classification developed by the Carnegie Foundation (available on the Carnegie Foundation website; see http://classifications.carnegiefoundation.org/lookup_listings/standard.php?key=783) and an educational profile variable used by the Carnegie Foundation to categorize colleges.

The educational profile variable used by the Carnegie Foundation is based on three components (variables). The first is the proportion of undergraduates who attend college part- or full time, where a full-time student is one who enrolled for 12 or more semester credits or quarter credits, or has 24 hours of instructional contact each week of a term (United States Department of Education, 2013). A second reflects the achievement characteristics of first-year, first-time students, defined using the ACT/SAT scores of the freshman class expressed in quintiles, for example the average ACT score in the upper one fifth of all colleges. The third characteristic represents the proportion of entering students who transfer in from another institution. A transfer-in student is defined as a student entering an institution for the first time but who previously attended another postsecondary institution as an undergraduate (United States Department of Education, 2013).

Although average ACT scores expressed in quintiles are used in constructing the educational profile variable, we also used a version of this variable as a standalone measure of selectivity (ACT score of the freshman class corresponding to the 25th percentile) because of its documented predictive power and because it showed a relatively small correlation with educational profile. Focusing on institutional characteristics like selectivity (average ACT score of the freshman class corresponding to the 25th percentile) and a college's educational profile, and ensuring that variation in these characteristics in our sample is consistent with that appearing on the Carnegie Foundation website, enhances the generalizability of our results.

Thirty-two of the 46 invited colleges (76%) agreed to participate and provided data for this study. Comparisons of colleges that did or did not participate provided no evidence of differences on the above institutional variables. For example, a two-sample *t*-test comparing average ACT mathematics scores corresponding to the 25th percentile for the 32 participating colleges and the 14 colleges not participating was not significant ($p > .05$). Similarly, a comparison of the two groups of colleges on the educational profile variable using a Kruskal-Wallis test for ranks was not significant ($p > .05$). The analyses reported below were based on 32 colleges in five states (Minnesota, Wisconsin, Iowa, North Dakota, and South Dakota).

The 32 colleges represent a purposive sample in the sense that they varied in ways we think enhanced the viability of generalizing to the target population. Shadish, Cook, and Campbell (2002) pointed out that purposive sampling does not ensure generalizability, but we think that the similarity of the 32 institutions to colleges in the United States on key indicators such as college preparation (numbers of students in the United States with ACT scores greater than or equal to 26 and SAT scores greater than or equal to 1200 per 1,000 high school graduates) and

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college accessibility (average tuition for public and private institutions in the United States) suggests generalizability of our results to a broader population.

The initial sample consisted of 13,337 students who completed at least 3 years of an NSF-funded, commercial, or UCSMP curriculum in a high school in Minnesota and subsequently enrolled in one of the 32 colleges in five contiguous states (Minnesota, Wisconsin, Iowa, North Dakota, and South Dakota). Participation was limited to students who completed at least 3 years of high school mathematics because students enrolling in college are generally assumed to have completed at least 3 years (Carnegie units) of high school mathematics, as reflected in the mathematics portions of the ACT and SAT (ACT, 2013; College Board, 2013).

All students had completed at least one mathematics course while they were enrolled in college. Fall 2002 was chosen as the first term to study college mathematics because it allowed for significant numbers of sampled students who had completed three or four years of the first published editions of NSF-funded curricula. Sampling students who enrolled in Fall 2003 essentially doubled the sample size. No evidence emerged that students enrolling in Fall 2002 differed systematically from those enrolling in Fall 2003, and students were pooled for the various analyses.

Data

Archival data were obtained from college records and reflected both high school and college information.

High school information. High school information was obtained from college records provided by the sampled postsecondary institutions and included titles of the high school mathematics courses a student had completed and the associated course grades for Grades 9–12, ACT mathematics score, years of high school mathematics completed, and demographic information. We adopted a two-step process to link each student to a curriculum cohort.

First, we contacted each of the 290 high schools (all in Minnesota) represented in our sample of students and obtained descriptions of their mathematics programs, including the mathematics courses offered and the mathematics textbooks used during the period the students were in high school (1998–2002). Thus, the response rate of our sample of high schools was 100%. Approximately 75% of the sampled students graduated from a high school in which a single curriculum (NSF-funded, commercial, or UCSMP) was offered, making it straightforward to assign these students to a curriculum cohort.

The remaining students (more than 3,000) graduated from a high school that offered two mathematics curricula (NSF-funded/commercial, NSF-funded/UCSMP, or commercial/UCSMP). Identifying the high school mathematics curriculum these students completed required a second step: cataloging the high school mathematics courses that appeared on the transcripts of these students and combining this with collateral information obtained from the high schools to assign a student to a curriculum cohort. Three project staff members completed Step 2.

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Two of these staff members collectively possessed more than 60 years of teaching experience with high school mathematics. In all cases, a student was assigned to a curriculum cohort after the three staff members agreed on the assignment.

College information. An archival data file containing available records for students who enrolled in college in Fall 2002 or Fall 2003 was obtained from the sampled postsecondary institutions. Longitudinal data for students who began college in Fall 2002 or 2003 were obtained through the Spring 2006 or 2007 college semesters, respectively. These records contained student college mathematics grades and courses completed across eight semesters and a student's declared college major. Information about institutions was obtained from each institution's website and from the Carnegie Classification of Institutions of Higher Education.

Variables

Predictors. The high school mathematics curriculum that a student completed was the key predictor. We opted to combine the NSF-funded curricula (Core-Plus, IMP, MMOW) in the analyses for two reasons. First, there are strong conceptual links among these curricula, including their adherence to a common set of standards in their construction. Second, we used MANOVA to look for differences among students who completed the Core-Plus, IMP, and MMOW curricula on high school variables including ACT mathematics score and high school mathematics GPA and, separately, on college variables including first college mathematics course grade and college mathematics GPA. The MANOVAs were statistically significant at $p < .05$ but produced squared canonical correlations of about .01, meaning that the particular NSF-funded curriculum accounted for little variation in the outcomes. Thus, we did not distinguish among individual NSF-funded curricula in our analyses.

Dummy-coded predictors were used to capture curriculum cohort—NSF-funded (1 = yes, 0 = no) and UCSMP (1 = yes, 0 = no)—and commercial served as the reference group, meaning, for example, that the slope for the NSF-funded predictor represented the difference on the outcome between the NSF-funded and commercial cohorts. Other student predictors were ACT mathematics score, female (1 = yes, 0 = no), and college major, which was coded using a modified version of the categories appearing in a United States Department of Education report (Chen, 2005): Life Sciences/Health Sciences and Services (1 = yes, 0 = no), Humanities/Fine and Performing Arts/Social Sciences/Education (1 = yes, 0 = no), Business (1 = yes, 0 = no), and Other (1 = yes, 0 = no); Engineering/Mathematics/Computer Science/Physical Sciences (STEM) served as the reference group. Number of years of high school mathematics completed (3, 4, or 5) also served as a student-level predictor. In constructing this variable, it was apparent that a small number of students completed more than one mathematics course in a single year. For these students, we coded the higher level course taken. The high school mathematics GPA variable was omitted from our final models because results with and without this predictor were quite similar.

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We coded ethnicity as a student predictor—African American (1 = yes, 0 = no), Asian American (1 = yes, 0 = no), and Hispanic (1 = yes, 0 = no)—so that white students served as the reference group. This pattern of coding ethnicity is common in regression analyses of achievement data (e.g., Attewell, Heil, & Reisel, 2011; Riegle-Crumb & Grodsky, 2010; Spenner et al., 2004). It is important to emphasize that the statistical results are identical regardless of which ethnic group serves as the reference group. Student predictors representing the interaction of ethnicity and curriculum cohort with respect to college mathematics outcomes also were constructed. These predictors allowed us to examine whether the relationship between college mathematics outcomes and ethnicity was related to curriculum (e.g., was one curriculum associated with a strengthening or weakening of this relationship).

Predictors at the institution level were enrollment, selectivity, financial profile, and educational profile. After examining institutional data for the 32 colleges, we found that three categories of educational profile characterized this feature of the institutions: (a) full-time/more selective/low transfer-in rate, (b) full-time/selective/high transfer-in rate, and (c) mixture of full-time and part-time/less selective/high transfer-in rate. Each institution's financial profile (public or private) also was coded.

Mathematics course taking was captured by creating a variable reflecting course difficulty level that matches the three-level scale used in Teitlebaum's (2003) analysis of the National Educational Longitudinal Study data but adds a fourth level:

- *Level 1*: This level includes developmental courses that should have been completed in high school. Students entering postsecondary study without the required or recommended high school mathematics background are typically directed to these courses, which are designed to provide a foundation for students to be successful in future mathematics courses and generally do not confer credits towards graduation.
- *Level 2*: This level includes courses that would be considered precalculus mathematics, including college algebra, precalculus, and finite mathematics.
- *Level 3*: This is the typical entry level for well-prepared high school students, who would start their college mathematics coursework with Calculus I.
- *Level 4*: This level is beyond a first course in calculus. Titles include Calculus II, differential equations, linear algebra, and multivariable calculus.

The difficulty-level variable was constructed by examining mathematics course descriptions at the 32 colleges and rating each course using the 4-point scale discussed above. The median number of mathematics courses offered by the colleges was approximately 23 and ranged from 3 to 47. Each course description was reviewed and rated independently by the three project staff members who examined high school mathematics courses. Initial levels of disagreement among

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raters never exceeded 1 point on the difficulty scale and constituted approximately 2% of the courses rated. The disagreements that did occur were usually confined to one or two courses at a few institutions (typically for difficulty level 2) and were subsequently resolved through discussion among the raters. Thus, the consistency of ratings used in our analyses was 100%.

Dependent variables. College mathematics course grades, which were based on different grading scales across institutions, served as dependent variables in several analyses. Seventy-five percent of the sampled institutions used a 12-point grading scale (A, A–, B+, . . . , F), 18.75% used a 7-point grading scale (A, AB, B, BC, C, D, F), and 6.25% used a 5-point grading scale (A, B, C, D, F). We wished to use data from all institutions in each analysis, and we examined the impact of differences in grading scale on our statistical modeling results to assess the credibility of doing so. Specifically, we performed sensitivity analyses using three versions of the grade variable (transforming all grades to a 5-point scale, transforming all grades to a 7-point scale, using the original grading scales) in our major analyses and compared the results. To create a 7-point scale based on a 12-point scale, we transformed grades such that A = A, A–/B+ = AB, B = B, B–/C+ = BC, C = C, D = D, F = F. To transform grades based on a 5-point scale to a 7-point scale, we assumed that the grades of A through F were earned on a 7-point scale (A = 7, A–/B+ = 6, . . . , F = 0) even though this was not always the case, meaning that students at institutions using a 5-point scale had no opportunity to earn an AB or BC grade. Throughout this process, we assumed that similar grades earned in similar courses at different institutions reflected similar proficiency. The statistical modeling results for the three versions of the grade variable were similar, and we report findings using the 7-point scale.

Two other dependent variables captured the likelihood that a student completed a college mathematics course with a difficulty equaling or exceeding that of Calculus I or that was developmental in nature. Each dependent variable was analyzed separately.

Important Methodological Issues

Selection bias. One potential source of bias appeared in high schools offering both NSF-funded and commercial curricula. The selection bias consisted of less mathematically proficient students being channeled into NSF-funded curricula at the beginning of high school (Harwell et al., 2009), creating preexisting differences among the cohorts. A selection bias of this nature would likely favor the commercial cohort as they moved into college mathematics coursework and disadvantage students who completed an NSF-funded curriculum. There may be a similar bias in high schools offering NSF-funded and UCSMP curricula or a commercial and UCSMP curriculum, but we have no evidence of this. In our study, this potential bias applies to 2,005 or 15% of the sampled students who graduated from one of 27 high schools in which NSF-funded and commercial curricula were offered.

To respond to the possibility of selection bias, we included statistical control

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variables in the analyses for which the curriculum cohorts differed significantly, including ACT mathematics scores. We note that previous results have suggested that mathematics proficiency is a fairly stable trait over time (e.g., Ding & Davison, 2005), a finding supported by Post et al. (2010), who found that ACT mathematics scores were essentially unaffected by high school mathematics curriculum. We also performed analyses to assess the sensitivity of our findings to such bias.

Bias due to missing data. An important data-based issue was the presence of missing college grade and difficulty level data. To respond to missing data and the associated possibility of biased inferences we turned to Little and Rubin's (2002) characterization of incomplete data through ignorable and non-ignorable response mechanisms. The data consisted of an $N \times P$ grid (where N represents the number of students and P represents the number of courses) in which 13,337 students had grades and difficulty levels for between 1 and 14 college mathematics courses. Because 87.3% of the students in our sample completed three or fewer college mathematics courses and 93.8% completed four or fewer courses, most of the cell entries in this grid were empty (incomplete or missing).

In the $N \times P$ data grid, most of the missing grade and difficulty values can be explained by the fact that college majors generally have requirements that specify the minimum number of mathematics courses students must successfully complete as part of their degree work and that students generally complete only required mathematics courses (Harwell et al., 2009). The inclusion of college major in the statistical modeling should minimize parameter estimation bias relative to ignoring missing grade and difficulty data (Little & Rubin, 2002).

We also examined our data for evidence of the impact of missing values for variables such as ACT mathematics score and years of high school mathematics. Data were missing relatively infrequently for most student variables (e.g., ACT mathematics scores were missing for 2.3% of students), although college major was missing more often (17% of the time). Missing college major values were overwhelmingly linked to students with Other majors, which included technologies, technicians, trades, military, and culinary arts. A consequence of missing data is that different analyses are based on different subsets of students. Extensive analyses comparing students for whom data were or were not available (e.g., ACT mathematics scores, college major) did not provide evidence of important differences in the pattern of significant results or the estimated magnitude of a statistically significant effect.

Other potential sources of bias. The fidelity with which high school mathematics curricula are implemented is an important methodological issue (O'Donnell, 2008) and applies equally to the implementation of NSF-funded, commercial, and UCSMP curricula. Although recent evidence suggests that fidelity is unrelated to student achievement (Krupa, 2011), we think that assessing fidelity of implementation continues to be both important and difficult, especially in a retrospective study like ours. One factor promoting fidelity for the NSF-funded curricula is that

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an earlier NSF-funded study (Minneapolis and St. Paul Merging to Achieve Standards Project [MASP]) provided more than 400 high school mathematics teachers from the Twin Cities area with an average of 120 hours of targeted professional development using NSF-funded curricula (see Harwell et al., 2007) between 1996 and 2001.

After completing the targeted professional development in NSF-funded curricula through MASP, the 400 high school mathematics teachers (representing almost all those teaching NSF curricula in the Twin Cities) collectively taught about 25,000 high school students during 1996–2002. Among the students in our sample who completed an NSF-funded curriculum, almost all graduated from a Twin Cities high school, meaning that they were very likely to have been taught by a MASP teacher. This suggests at least moderate fidelity of implementation of NSF curricula, although significant variation is likely even among MASP teachers (Tarr et al., 2008).

The alignment (or lack thereof) between high school mathematics curricula and college mathematics placement tests was also a potential source of bias in its impact on college placement test scores and advising students on which college mathematics course to enroll in initially. Available evidence suggests that the strongest alignment is between a commercial curriculum and material likely to appear on college mathematics placement tests and, thus, is most likely to benefit students who completed a commercial curriculum (Norman, Medhanie, Harwell, Anderson, & Post, 2011). There does not appear to be any available evidence of this effect for the UCSMP curriculum.

Another important issue in estimating the relationship between high school mathematics curricula and college mathematics achievement and course taking is variation in the quality of teaching within and between institutions and institutional assessment practices (Wayne & Youngs, 2003). However, we had no data capturing these factors.

Cross-curricular contamination in a high school offering two curricula could also be a source of bias. For example, teachers teaching a commercial curriculum and those teaching an NSF-funded curriculum in a high school could, as a result of professional or social interactions, actively incorporate features of the other curriculum that they thought useful into their teaching and bias the results. Student transience (students transferring to high schools that use a different curriculum) and variation in high school and class size, which have been shown to affect student achievement (Nye, Hedges, & Konstantopoulos, 1999), also represented potential sources of bias. Unfortunately, we did not have data to empirically test for these biases.

In summary, the potentially damaging effects of different sources of bias on our inferences represent important methodological challenges. Although we were unable to respond to all threats, we think that our response to several threats, using control variables and sensitivity analyses, is consistent with recommended practice (What Works Clearinghouse, 2008), and, combined with evidence that key threats were unlikely to seriously distort our findings, allows us to credibly use our results to answer our research questions.

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Data Analysis

We used descriptive analyses—two-level (students within institutions) and three-level (longitudinal grade and difficulty level data within students within institutions) multilevel models—to capture the hierarchical nature of the data and the associated dependency in outcomes within students and institutions (see Raudenbush & Bryk, 2002). Dependencies in college mathematics outcomes are likely among students with similar pre-college backgrounds (e.g., high school experiences, academic background) who enroll in the same institution (Spenner et al., 2004; Whitt, Pascarella, Nesheim, & Marth, 2003).

We did not include high school as a level in our analyses because we did not have sufficient data to support a model that also included student- and institution-level predictors. To assess the possible impact of not including high school in our models, we fitted an unconditional (no predictors) cross-classified random effects model with high schools and colleges as factors (Raudenbush & Bryk, 2002) to our cross-sectional outcomes. There was no evidence of important differences between high schools for these outcomes. For example, the cross-classified random effects model fitted to Calculus I grades showed that the intraclass correlation for high schools (i.e., correlation among Calculus I grades of two students from the same high school who attended different colleges) was .01.

We reformulated the semester variable (1 through 8) to capture mathematics courses completed (first mathematics course completed, second mathematics course completed, and so on) to facilitate interpretation. Most students completed 3 or fewer college mathematics courses, but a few completed up to 14 courses, raising concerns that these latter values might reflect outliers in the data analyses and disproportionately impact the findings (Langford & Lewis, 1998). To respond to this possibility, we limited the number of mathematics courses considered in our analyses to seven. For the few students who completed more than seven mathematics courses, the course beyond the seventh with the highest difficulty level was used as the seventh course and the remainder of the course data was excluded.

To construct a three-level model, we generally followed the steps recommended by Bliese and Ployhart (2002) to identify the explanatory model that provided the best fit to the data. This led us to divide the analyses into two phases for each outcome variable that involved (a) identifying a between- or within-student model with predictors and an error structure consistent with the data and determining if slopes of predictors should be fixed (not allowed to vary across students/institutions) or random (allowed to vary across students/institutions) effects and (b) performing statistical tests of the fixed effects.

To control for compounding of Type I error rates and ensure that the overall error rate equaled that specified, we used an adjusted error rate attributed to Sidak (1967). The adjusted error rate was computed as $\alpha' = 1-(1-\alpha)^{1/m}$ where α is the unadjusted Type I error rate, *m* represents the number of statistical tests within an analysis phase for a given outcome, and α' is the adjusted Type I error rate. We chose an overall $\alpha = .15$ for the statistical tests of fixed effects associated with each outcome variable. This practice is consistent with the frequently recommended

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familywise error rate strategy for controlling Type I error rates without severely dampening statistical power (Keselman, Cribbie, & Holland, 2002). Preliminary analyses showed that the financial profile variable had little explanatory power, so it was removed from the model.

Grades. For cross-sectional analyses of the Calculus I grade data (Research Question 1a) we fitted a two-level model (students within institutions). Research Questions 1b and 1c were answered by fitting a three-level model (longitudinal grades within students within institutions). A review of the literature for college mathematics grades did not suggest a theoretical model for the expected pattern of student grades over time, and an empirical examination of the grade and difficulty level trajectories led us to use models in which the patterns over courses were linear. For longitudinal analyses of grades, we assumed that within-student error terms followed an autoregressive structure such that the error associated with the grade of the ith student in his or her pth college mathematics course depended on the error associated with the grade of his or her $(p-1)$th mathematics course (Raudenbush & Bryk, 2002).

Following the advice of Biesanz, Deeb-Sossa, Papadakis, Bollen, and Curran (2004), we rescaled the predictors in the within-student model to answer questions of interest. Specifically, mathematics courses completed was rescaled so that 0 represented the last college mathematics course completed, 1 represented the next to last college mathematics course completed, and so on. Similarly, the 4-point difficulty level variable was rescaled so that 0 represented the most difficult college mathematics course a student completed, which was almost always a student's last mathematics course. The rescaling allowed us to interpret the intercept estimated for each student in the three-level model as the expected grade in a student's last and most difficult college mathematics course and to use multilevel analyses involving student intercepts to answer key research questions.

Because the difficulty level variable had an ordinal scale of measurement, the models for grade were also fitted using three dummy-coded predictors to capture the information in the difficulty level variable. The multilevel findings using difficulty level as a predictor in its original form and those obtained using dummy-coded predictors to represent difficulty level produced the same pattern of findings, and the results for difficulty level in its original (4-point) form are reported.

Predictors for the student and institution models were grand-mean-centered so that interpretations of slopes are in terms of a student or institution whose values on these variables are average. An average student is one whose value for these predictors equals the grand mean of these variables. For example, the estimated grand mean for ACT mathematics scores was 24, and an average student is one whose ACT mathematics score equals this value. Frequently, there are no actual cases with values for the predictors exactly equal to the estimated grand means of these predictors, and the phrase *average student* is for descriptive purposes.

Difficulty levels. To answer our research questions related to the difficulty level outcome, the original 4-point difficulty level variable was dichotomized to create

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two separate dependent variables. One dependent variable for difficulty level was defined such that 1 indicated that a student completed Calculus I or beyond and 0 otherwise. Research Questions 2a and 2b were answered by fitting generalized hierarchical proportional odds models to these data (see Hosmer & Lemeshow, 2000). The predictors in these models were the same as those for the grade data except that difficulty level was not a predictor in the within-student model and an autoregressive error structure for difficulty levels was assumed.

A second dependent variable for difficulty was defined such that 1 indicated that a student completed a developmental mathematics course and 0 otherwise. Research Question 2c was answered by fitting generalized hierarchical proportional odds models to this cross-sectional outcome. The predictors in this model were the same as those used for analyses of grades.

Model checking revealed some evidence of departures from underlying assumptions, primarily for a subsample of two to three institutions whose residuals did not appear to be consistent with the assumption of multivariate normality. Refitting our models without this subsample (whose composition varied over dependent variables) produced the same general pattern of findings as those using all of the institutions, and the results reported are based on the full sample of institutions. Also, no evidence of collinearity emerged in preliminary analyses. Multilevel analyses were performed with the R 2.10.1 lme4 package (R Development Core Team, 2012), and full maximum likelihood (Raudenbush & Bryk, 2002) was used to estimate parameters.

Results

Next, we present results for a series of descriptive and multilevel analyses of the college mathematics grade and difficulty level data. The descriptive analyses reflect patterns in the data, and the results of the multilevel analyses provide evidence about the research questions.

Descriptive Analyses

Among postsecondary institutions, the average enrollment based on information provided on each institution's website was 4,953 (SD = 6,577), but there was substantial variability across institutions as summarized using quartiles (Q1 = 1,795, Q2 = 2,449, Q3 = 4,201). Samples from each institution also varied, with seven colleges providing less than 100 students and seven providing more than 500. Table 1 reports various summary statistics for students by educational profile. These statistics suggest that the distribution of students across the curriculum cohorts was approximately equal within each of the three categories of the educational profile variable. For example, the percentage of full-time students in more selective institutions who completed the NSF, UCSMP, or commercial curricula in high school were 55.8%, 57.3%, and 56.7%, respectively.

Table 1 also shows that far fewer (N = 301) but proportionally more African American students (73.1%) were in more selective institutions compared to white

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Table 1
Summary Statistics for Curriculum Cohort, Female, Ethnicity, and College Major by Educational Profile of the Sample Postsecondary Institutions

		Educational profile					
		Full-time/more selective/ low transfer-in rate[a]		Full-time/ selective/ high transfer-in rate[b]		Mixture of full-time and part-time, less selective/ high transfer-in rate[c]	
		N	%	*N*	%	*N*	%
High school mathematics curriculum	NSF-funded	971	55.8	754	43.3	16	0.9
	UCSMP	1,292	57.3	950	42.1	12	0.5
	Commercial	4,598	56.7	3,486	43.0	22	0.3
Female	Yes	3,689	50.2	3,633	49.5	22	0.3
	No	3,691	61.7	2,254	37.7	37	0.6
Ethnicity	Asian	843	78.6	229	21.4	0	0.0
	African American	301	73.1	111	26.9	0	0.0
	Hispanic	135	63.1	79	36.9	0	0.0
	White	5,993	52.6	5,345	46.9	57	0.5
College major	Life Sciences	1,144	54.7	949	45.3	0	0.0
	Humanities	2,757	60.2	1,819	39.8	0	0.0
	Business	849	38.3	1,368	61.7	0	0.0
	Other	67	21.3	247	78.7	0	0.0
	STEM	1,247	66.6	3,486	43.0	0	0.0

[a] Full-time/more selective means at least 80% of students attend school on a full-time basis and average ACT score is in upper one fifth of all colleges. Low transfer-in rate means less than 20%. [b]Full-time/selective/high transfer-in rate means at least 80% of students attend school on a full-time basis, average ACT score is in middle two fifths of all colleges, and transfer-in rate of at least 20%. [c]Less selective means average ACT score is in bottom two fifths of all colleges.

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Table 2
Summary Statistics by Educational Profile

		Educational profile								
		Full-time/more selective/ low transfer-in rate			Full-time/ selective/ high transfer-in rate			Mixture of full-time and part-time, less selective/ high transfer-in rate		
		M	*SD*	*N*	*M*	*SD*	*N*	*M*	*SD*	*N*
ACT Math 25th %[a]		25.16	4.62	7,239	22.50	4.22	5,728	22.05	4.07	57
Grade first college math course		2.82	1.02	7,225	2.61	1.15	5,693	2.55	1.19	51
Grade last college math course		2.77	0.89	6,915	2.63	1.06	5,566	2.66	1.24	59
Average grade by difficulty level[b]	1 (Dev)	2.31	1.19	1,513	2.32	1.19	1,286	2.06	1.05	32
	2 (Coll Alg)	2.49	1.12	6,944	2.53	1.17	6,429	2.46	1.10	76
	3 (Calc I)	2.67	1.06	8,035	2.71	1.08	4,142	3.00	0.88	52
	4 (> Calc I)	2.89	1.01	8,029	2.92	0.97	2,976	3.21	0.75	28
		%		*N*	%		*N*	%		*N*
First college math course (proportion in each difficulty level)	1 (Dev)	9.2		678	12.8		747	23.7		14
	2 (Coll Alg)	42.8		3,145	63.9		3,713	50.8		30
	3 (Calc I)	33.7		2,477	19.9		1,158	25.4		15
	4 (> Calc I)	14.2		2,045	3.4		197	0.0		0
Last college math course (proportion in each difficulty level)	1 (Dev)	4.6		320	8.7		488	6.8		4
	2 (Coll Alg)	38.4		2,689	60.6		3,398	59.3		35
	3 (Calc I)	25.2		1,766	17.0		951	23.7		14
	4 (> Calc I)	31.8		2,225	13.7		768	10.2		6

Note. Dev = developmental (difficulty level 1, courses that should have been taken in high school), Coll Alg = college algebra/precalculus (difficulty level 2), Calc I = Calculus I (difficulty level 3), > Calc I = beyond Calculus I (difficulty level 4).
[a]Scores correspond to the 25th percentile and were taken from institution websites. [b]The sample sizes represent the total number of students who completed a course at that level; students who took courses at different difficulty levels were counted in the sample size for each level.

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students (5,993 and 52.6%, respectively). The number of students in our sample who were full-time or part-time students at less selective institutions with high transfer-in rates was quite small, and inferences based on the reported percentages should be made cautiously.

Table 2 reports summary statistics for ACT mathematics scores corresponding to the 25th percentile, average grade earned in each difficulty level, and the proportion of students in each difficulty level for the first and last college mathematics courses by educational profile. These results suggest that students in more selective institutions tended to be more proficient in mathematics and to enroll in higher level mathematics courses, compared to students in less selective institutions. For example, 33.7% of full-time students in more selective institutions with low transfer-in rates began their mathematics coursework with Calculus I, whereas 19.9% of full-time students in less selective institutions with high transfer-in rates did so.

Table 3 reports summaries for students' ACT mathematics score, first and last college mathematics course grades, and the difficulty level of students' first and last college mathematics courses by curriculum cohort across institutions. There was little difference in effect sizes between the UCSMP and commercial curriculum cohorts.

Bivariate correlations among student variables were also computed. The largest correlation was between grade earned in a student's first and last college mathematics courses (.76), followed by the correlation between ACT mathematics score

Table 3
Summary Statistics for Student ACT Mathematics Scores, Grades, and Difficulty Level of First and Last College Mathematics Courses by Curriculum Cohort

	High school mathematics curriculum								
	NSF-funded			UCSMP			Commercial		
	M	*SD*	*N*	*M*	*SD*	*N*	*M*	*SD*	*N*
ACT Math	22.30	4.50	1,705	24.60	4.30	2,190	24.20	4.70	7,933
Grade first college math course	2.61	1.12	1,690	2.74	1.08	2,203	2.74	1.09	7,895
Grade last college math course	2.62	0.99	1,637	2.73	0.97	2,122	2.71	0.97	7,619
Difficulty level first college math course	2.11	0.80	1,728	2.48	0.87	2,243	2.41	0.87	8,037
Difficulty level last college math course	2.50	1.10	1,655	2.86	1.14	2,143	2.80	1.16	7,696

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and number of years of high school mathematics (.46); all other correlations were less than .30 and thus reflect moderately weak to weak relationships.

Multilevel Analyses

Multilevel models were used to study the relationship between curriculum cohort and college mathematics achievement and course taking. These models reflect data obtained at multiple levels of a data hierarchy and include both fixed effects and random effects. The grade and difficulty level variables were analyzed separately, and the findings presented below are organized by research question. Predictors representing the interaction of ethnicity and curriculum with respect to college mathematics outcomes were initially included in the models but were not statistically significant, meaning that the relationship between college mathematics outcomes and ethnicity was unrelated to the high school mathematics curriculum a student completed.

Research Question 1a: College Calculus I grades. Two-level (students within institutions) cross-sectional fixed effects results for Calculus I grades are reported in Table 4. These analyses estimated the effect of each predictor on the outcome via a slope (*B*) and tested each slope using a *t*-test. Table 4 also reports standardized slopes (*Stand B*) for statistically significant effects to enhance interpretation.

The largest effects in Table 4 were for college major, with an average student in STEM (i.e., a student whose values for the predictors in the model, such as ACT math, equaled the grand mean of those variables) earning a grade that was 0.324 units (on a 7-point scale) higher than an average student whose major was categorized as Other. Equivalently, an average STEM major earned a grade that was 0.334 standard deviations higher than students whose college major was Other. Average male students earned a Calculus I grade that was 0.12 units (0.13 *SD*) higher than that earned by average female students. ACT mathematics scores and years of high school mathematics were also significant predictors of Calculus I grades. However, ethnicity was not a significant predictor, meaning that African American and white students had, with the other predictors held constant, similar Calculus I grades. The predictors representing Asian American and Hispanic students, and the institutional predictors, were not significant.

We performed a sensitivity analysis for these findings to assess the impact of specific features of the data and the model on the results. Findings that do not appear to be overly influenced by these specific features provide evidence that our results are not seriously distorted by threats to inferences such as selection bias.

One sensitivity analysis was performed in response to concerns about the impact of less mathematically proficient students being channeled into NSF-funded curricula at the beginning of high school (Harwell et al., 2009). We conditioned the analysis of college Calculus I grades on the difficulty level of a student's first college mathematics course (determined by project staff as described earlier) by refitting our model for students who began their college mathematics course taking

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Table 4
Multilevel Cross-sectional Fixed Effects Results for College Calculus I Grades

Between-student model				
Effect	*B*	*SE*(*B*)	*t*-value	*Stand B*
Intercept	2.588	0.042	61.72	—
NSF-funded	–0.004	0.048	n.s.	—
UCSMP	– 0.02	0.039	n.s.	—
Female	– 0.120	0.031	– 3.83	–0.130
African American	0.061	0.105	n.s.	—
Asian	– 0.029	0.057	n.s.	—
Hispanic	0.122	0.141	n.s.	—
Life Sciences	– 0.007	0.042	n.s.	—
Humanities	– 0.217	0.043	– 5.03	–0.224
Business	– 0.050	0.051	n.s.	—
Other	– 0.324	0.081	– 3.99	–0.334
ACT Math	0.074	0.004	16.56	0.077
Years HS Math	0.169	0.026	6.58	0.175
Between-institution model				
Effect	*B*	*SE*(*B*)	*t*-value	*Stand B*
Enrollment	0.00	0.00	n.s.	—
Selectivity[a]	0.023	0.023	n.s.	—
Educational Profile	–0.099	0.112	n.s.	—

Note. Analyses were based on data from 30 postsecondary institutions and 3,649 students. *Stand B* = standardized effect sizes. n.s. = not significant at $\alpha' = 1 - (1 - \alpha)^{1/m} = .00101$, where $\alpha = .15$ is the unadjusted Type I error rate, m = number of statistical tests (16 fixed effects), and α' is the adjusted error rate.

[a]ACT score of freshman class corresponding to the 25th percentile.

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with a course of similar difficulty. This strategy helped to minimize the possible effects of channeling. This resulted in two sets of analyses that corresponded to the second and third difficulty levels, College Algebra and Calculus I. Students who began their college mathematics course taking with a course of difficulty level 1 were omitted from these sensitivity analyses because there were a small number of cases, and those who began with a course of difficulty level 4 (Calculus II or beyond) never took Calculus I in college. There was agreement between the above findings and those produced by the sensitivity analyses for the effect of curriculum cohort on Calculus I grades. Specifically, the same pattern of significant and nonsignificant effects emerged, providing evidence that the main results were not overly influenced by selection bias attributable to channeling.

We also refitted the Calculus I grade model using a sample of students known to have graduated from a high school offering a single mathematics curriculum, and, separately, to a sample of students who graduated from a high school offering two different mathematics curricula. The pattern of significant and nonsignificant results was the same and provided evidence that our results were not sensitive to whether high schools offered one or two curricula.

Research Questions 1b and 1c: Grades across college courses. Three-level analyses of college mathematics course grades are reported in Table 5. Curriculum cohort was not a significant predictor of grade earned in a student's last college mathematics course (NSF-funded slope = 0.022, n.s.; UCSMP slope = –0.038, n.s.), but other student variables were. For example, average African American and Asian American students earned a grade in their last college mathematics course that was 0.28 units (0.37 *SD*) and 0.18 units (0.23 *SD*), respectively, below that earned by average white students. Average female students tended to earn a grade that was 0.19 units (0.25 *SD*) lower than males in their last mathematics course. Overall, the between-student intercept model reported in Table 5 for last college mathematics grade accounted for 12% of the variance compared to the unconditional model.

The within-student model in Table 5 shows that the more mathematics courses a student completed, the more difficult these courses tended to be (B = 0.403 or 0.52 *SD*). Student grades increased slightly per mathematics course completed, even though courses typically increased in difficulty (B = –0.079 or –0.102 *SD*). Mathematics courses completed was coded so that the last course completed was coded 0, the next to last course was coded 1, and so on. This pattern is likely attributable to the fact that students who completed more college mathematics courses were more mathematically proficient than students who completed fewer mathematics courses before entering college. For example, the average ACT mathematics score for students who completed one nondevelopmental course (almost always college algebra) was 23.60 (SD = 4.28); for students who completed two or more nondevelopmental mathematics courses (usually College Algebra and Calculus I), the average ACT mathematics score was 25.16 (SD = 4.64). This difference in means is statistically significant ($p < .05$) and suggests that students who completed more college mathematics courses were more mathematically proficient than their peers.

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Table 5
Multilevel Results for Longitudinal Grade Data

Within-student model				
Effect	*B*	*SE*(*B*)	*t*-value	*Stand B*
Intercept	2.682	0.040	67.21	—
Linear slope	–0.079	0.014	–5.61	–0.102
Difficulty slope	0.403	0.016	25.51	0.521
Between-student model				
Effect	*B*	*SE*(*B*)	*t*-value	*Stand B*
Intercepts				
NSF-funded	0.022	0.029	n.s.	—
UCSMP	–0.038	0.026	n.s.	—
Female	–0.193	0.021	–9.33	–0.253
African American	–0.284	0.065	–4.39	–0.373
Asian	–0.176	0.039	–4.57	–0.231
Hispanic	–0.072	0.083	n.s.	—
Life Sciences	0.046	0.034	n.s.	—
Humanities	0.150	0.030	4.95	0.197
Business	0.137	0.034	4.00	0.180
Other	–0.154	0.059	–2.61	– 0.202
ACT Math	0.065	0.003	23.78	0.085
Years HS Math	0.062	0.016	3.77	0.081
Linear Slopes				
NSF-funded	–0.032	0.017	n.s.	—
UCSMP	0.00	0.014	n.s.	—
Female	0.021	0.012	n.s.	—
African American	0.085	0.031	2.71	0.111

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Asian	0.041	0.019	n.s.	—
Hispanic	0.109	0.052	n.s.	—
Life Sciences	–0.054	0.019	–2.88	–0.071
Humanities	–0.022	0.015	n.s.	—
Business	–0.009	0.022	n.s.	—
Other	–0.042	0.031	n.s.	—
ACT Math	0.002	0.001	n.s.	—
Years HS Math	0.013	0.009	n.s.	—

Between-institution model

Effect	*B*	*SE*(*B*)	*t*-value	
Intercepts				
Enrollment	0.00	0.00	n.s.	—
Selectivity	–0.001	0.023	n.s.	—
Educational profile	–0.052	0.113	n.s.	—
Linear Slope				
Enrollment	0.00	0.00	n.s.	—
Selectivitya	0.012	0.007	n.s.	—
Educational profile	–0.003	0.035	n.s.	—

Note. Analyses were based on data from 32 postsecondary institutions and 10,288 students. *Stand B* = standardized effect sizes. n.s. = not significant at $\alpha' = 1 - (1 - \alpha)^{1/m} = .00101$, where $\alpha = .15$ is the unadjusted Type I error rate, m = number of statistical tests (33 fixed effects), and α' is the adjusted error rate. Reported coefficients were statistically significant.
[a]ACT score of freshman class corresponding to the 25th percentile.

Similarly, one half of STEM majors (about 29% of the sample) completed Calculus II or a more difficult course and earned an average grade of 2.77 in these classes whereas for non-STEM majors, one half completed college algebra or a more difficult course and earned an average grade of 2.71 in these classes. Thus, taking more mathematics was associated with more difficult courses in which students earned slightly higher grades. Overall, 5.5% of the variance within students in the model in Table 5 was accounted for, a relatively modest effect.

Curriculum cohort was not a significant predictor of students' linear grade trajectories (pattern of grades over mathematics courses). Table 5 shows that the largest

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effect in the slope models involved African American students (0.085 units, 0.11 *SD*), meaning that an average African American student's linear grade trajectory increased at a slower rate than that of an average white student. None of the institution predictors were significant. Difficulty level slopes varied significantly, meaning that the difficulty of mathematics courses completed varied across students but none of the predictors in this model were significant.

We performed a sensitivity analysis for Table 5 results by conditioning the longitudinal analysis of college mathematics grades on the difficulty level of a student's first college mathematics course. Once again, we refitted our multilevel models to the grade data for students who began their college mathematics course taking with a course of similar difficulty, and there was strong agreement between the main findings and those produced by the sensitivity analyses, including no significant impact of curriculum on grades.

We also refitted the longitudinal grade models to subsamples of students known to have graduated from a single or a dual curriculum high school. The pattern of the results was similar to those reported in Table 5 with one exception: For students from high schools offering both NSF-funded and commercial curricula, the NSF-funded curriculum (1 = yes, 0 = no) by mathematics course completed slope was significant (–0.06), meaning that the linear grade trajectories of the NSF-funded cohort increased at a slightly faster rate than that of the commercial curriculum cohort. To explore this pattern further we refitted the main longitudinal grade model after adding a predictor indicating whether a student graduated from a single or a dual curriculum high school and found that the pattern of significant and nonsignificant results was the same as that reported in Table 5.

Research Questions 2a and 2b: Course difficulty levels. The results of fitting a three-level model to a binary outcome variable, in which 1 indicated a course difficulty equal to or greater than Calculus I and 0 otherwise, showed that none of the predictors were statistically significant. Thus, curriculum cohort, ethnicity, and gender were not related to the difficulty level of a student's last college mathematics course or to whether they completed Calculus I or a higher level college mathematics course.

Research Question 2c: Developmental course taking. Our data indicated that overall, 22%, 12%, and 10% of the sampled students who completed an NSF-funded, commercial, or UCSMP curriculum, respectively, began their college mathematics with a developmental course. To explore this further, we fitted a two-level (students within institutions) model to a binary outcome in which 1 indicated a student completed a developmental mathematics course in college and 0 otherwise. Fitting this model with the predictors appearing in Table 5 showed that curriculum cohort was a significant predictor of the likelihood that a student completed a developmental mathematics course in college. An average student in the NSF-funded cohort was about 1 1/3 times as likely to initially take a developmental mathematics course in college than to take a more difficult course, compared

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to students in the commercial curriculum cohort. Equivalently, the model-implied probability (Raudenbush & Bryk, 2002) that a student who completed an NSF-funded curriculum will take a developmental mathematics course was .58.

Discussion

Advances in learning theory and the acknowledgement of students' interests and abilities increasingly play significant roles in the conceptualization of the mathematics curricula designed for school-age students. The NSF-funded and UCSMP programs studied here were guided by the 1989 NCTM Standards and can be thought of as reflecting the best practices in the field of teaching and learning of school mathematics at the time (NCTM, 1991). However, there have been relatively few studies comparing the impact of different curricula on students' mathematics proficiency, especially among college students.

This study contributes to the empirical research base by characterizing the extent to which high school mathematics curricula prepare college-bound students for college mathematics. Our findings can be organized into three complementary categories. First, our results suggest that NSF-funded, commercial, and UCSMP curriculum prepared students who began with a nondevelopmental course in college (e.g., College Algebra or Calculus I) equally well in terms of longitudinal mathematics achievement and course-taking patterns across a range of colleges, including both public and private institutions. Collectively, these results suggest that the largely anecdotally based criticisms that NSF-funded curricula leave students unprepared to succeed in college calculus and more difficult courses are unsubstantiated. The results also extend earlier findings by Schoen and Hirsch (2003), Harwell et al. (2009), and Post et al. (2010), who examined similar data from a single postsecondary institution.

In examining our findings, the absence of statistical significance for the high school mathematics curriculum variable raises the question of whether there was sufficient statistical power to detect this effect. Prior to conducting our study, a power/sample size analysis that focused on testing the curriculum effect for cross-sectional and longitudinal data was done using the Optimal Design software (Raudenbush, Liu, Spybrook, Martinez, & Congdon, 2006). Specifically, we projected that a sample size of 35 postsecondary institutions and 5,000 students would allow us to detect a standardized curriculum effect of .35 (a medium effect following Cohen, 1988) with a power exceeding .80 in most analyses. Although the final number of sampled students generally exceeded 10,000, the number of institutions in our sample was 32. Juxtaposed against the slightly smaller than anticipated number of postsecondary institutions is that *p*-values for tests of the estimated curriculum effect for the various outcomes typically exceeded .30, suggesting that the lack of statistically significant curriculum effects is unlikely to be attributable to inadequate power.

Second, our analyses indicate that students who completed an NSF-funded curriculum were more likely to enroll initially in a developmental mathematics

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course in college, compared to students who completed a commercial or UCSMP curriculum. This result may of course be due to differences in the ability of these curricula to prepare students for courses situated at difficulty level 2 (college algebra), but other explanations are also plausible. For example, these results may reflect a bias in which less proficient students tend to be channeled into an NSF-funded curriculum in high schools that also offer a commercial curriculum, the differential advising NSF-funded curriculum students may receive as to which college mathematics course to enroll in initially, a relative lack of alignment between NSF-funded curricula and college mathematics placement tests, or some combination of these (Norman et al., 2011). We also found that students from different curricula were equally likely to finish their college mathematics course taking with a course of similar difficulty, which we think is important.

The study by Norman, Medhanie, Harwell, Anderson, and Post (2011) highlights the importance of both high schools and colleges participating in serious efforts at realignment. For example, high school curricula should ideally include content appearing in college placement tests, and colleges need to be aware of the content differences of the various high school curricula and ensure that placement tests reflect the content and approaches of these newer curricula. However, the locally developed nature of many college mathematics placement tests, and the absence of evidence that these tests predict subsequent student mathematics achievement (Medhanie, Dupuis, LeBeau, Harwell, & Post, 2012), significantly complicates this task. Still, the introduction of the Common Core State Standards for Mathematics (National Governors Association Center for Best Practices & Council of Chief State School Officers, 2010) offers directions for the changes that both high schools and postsecondary institutions must make to reach alignment with the standards. The Common Core State Standards for Mathematics have parallel tracks, one labeled traditional and the other designated as integrated, with the latter seemingly designed to accommodate alternative approaches to high school mathematics.

Third, the presence of a grade trajectory for African American students that did not increase as fast as that of white students, and differences between African American and white students in the grade earned in their last college mathematics course, are consistent with literature indicating that the achievement gap in mathematics in high school continues into college. Spenner, Buchmann, and Landerman (2004) argued that much of the gap in mathematics achievement between African American and white students in their first college semester can be explained by differences in educational experiences beginning in early childhood, as well as family backgrounds of students prior to college enrollment (see also Treisman, 1992).

Spenner et al. (2004) also suggested that institutional climate, reflecting the extent to which campus environments (e.g., classroom, dormitory) are perceived as welcoming and inclusive or hostile and exclusive, may impact the mathematics achievement gap between African American and white students. Spenner et al. did not have data to test this assertion empirically, and, unfortunately, neither did we. However, our finding of no interaction between student ethnicity and curriculum and our outcomes provide evidence that enrollment in new and different high

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school mathematics curricula, in and of itself, is unlikely to significantly impact the mathematics achievement gap.

In summary, our results offer a clear take-home message: There is no single road to success in college mathematics. Alternate conceptualizations of the content, varied instructional approaches, and varied assessment practices resulted in similar success in mathematics for thousands of students who began college mathematics with college algebra or a higher level mathematics course. These results can be seen as a contribution to understanding the efficacy of these best practices for the mathematics classroom. The NSF-funded curriculum students who began their college mathematics with college algebra or a more difficult course showed college achievement patterns and course difficulty levels that equaled that of students who completed other curricula in high school. These results provide support for the instructional approaches and organization of the content domains reflected in NSF-funded curricula.

For postsecondary institutions, our findings suggest that the mathematics curriculum a student completed in high school is unlikely to predict differential success in college mathematics. This should inform the development and modification of college mathematics placement tests, initial course enrollment recommendations, and perhaps even the design of college courses.

Limitations

There are important limitations of our study that should be considered in examining our results. One is evidence of a selection bias connected to student enrollment in an NSF-funded curriculum (less proficient students tended to be channeled into an NSF-funded curriculum in high schools also offering a commercial curriculum). The use of control variables in our main analyses and the associated sensitivity analyses provide evidence that this bias does not overly influence our findings, but it is impossible to be sure. It is also possible that a school-level bias exists in which schools with lower levels of student achievement are more likely to experiment with the adoption of NSF-funded curricula.

A second limitation is fidelity of implementation. A portion of the high school mathematics teachers were involved in a targeted NSF-funded professional development program of 120 hours or more over a 2-year period. Our evidence suggested that they implemented the NSF-funded curricula with a reasonable amount of fidelity (Harwell et al., 2009; Post et al., 2010). This, however, does not eliminate concerns about the potentially important biasing effects of this factor on a wider sample of teachers (O'Donnell, 2008) and applies to the implementation of all curricula, commercial and UCSMP included. We recognize that fidelity of curricula implementation is a key factor in attempts to assess the viability of a new mathematics curriculum and that fidelity is a complex concept that can be measured in more than one way (e.g., content coverage, pedagogical approach, interview, classroom observation).

We also acknowledge that the degree of departure from more traditional curricular and pedagogical approaches may be significantly larger and more complex for

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teachers of NSF-funded curricula than for teachers of the other two curricula. In our experience, implementing NSF-funded curricula in ways that accurately reflect the authors' intent requires a particularly intense effort by teachers that requires substantial time and resources. Whether this leads to implementing NSF-funded curricula with more or less fidelity is at this point unclear.

Other important limitations include purposive sampling of colleges, which does not ensure that inferences can be generalized beyond our sample, and a quasi-experimental design that places limits on the causal inferences that can be drawn. The impact of missing data and student mobility and the possibility of important differences within and between institutions in the quality of mathematics instruction and variation in assessment practices should also be kept in mind in examining our results.

A more general limitation is that the analyses conducted here were based on NSF-funded first edition materials, meaning that our study would be difficult to replicate using second edition materials, as these have only recently been released after renewed funding by NSF. Assuming that a replication would require 4 years of high school performance data followed by 4 years of college mathematics achievement and course-taking data, an additional 8 years would be needed to allow the kind of curriculum comparisons reported in this study. Of course, there is no guarantee that the new editions released in 2008–2009 are more (or less) effective than the first editions, but it is reasonable to posit that the authors of the second edition materials have learned a great deal after a decade of implementation of their first edition materials.

What Now?

The general finding that the high school mathematics curriculum students complete has no significant relationship with their college mathematics achievement in college algebra or a more difficult course raises the question of what should happen next in this line of research. Our findings suggest a need for studies identifying the factors associated with the slower rate of increase in mathematics grades across courses taken by African American students and the nature of the differences between African American and white students' grades in their last college mathematics course. Information needed to address this important topic was not available to us in this study. Moreover, our sense is that future work might be well served by employing a mixed methods research design that draws on both qualitative and quantitative methodologies. There is also a need to explore the disproportional representation of college students who completed an NSF-funded curriculum in developmental mathematics courses.

Still, it is essential to consider the implications of the general finding that the various curricula are preparing students equally well for college mathematics. Is this license to continue current instructional practices, which are largely defined by a commercial curriculum with less than stellar student achievement and retention results over the past several decades? Should the research focus shift to other facets related to how students who completed various high school mathematics

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curricula perform in the mathematics classroom, such as tenacity, motivation, teacher reactions and insights, problem-solving ability, and fidelity of implementation? Should the research community continue to examine more closely differential achievement patterns for specific mathematical topics in the various high school and college curricula, and should the focus be on employing longitudinal studies rather than cross-sectional studies? Should we try to answer all the questions above?

Is there an argument for using NSF-funded curricula based on their links to the NCTM-endorsed Standards and the national goal of having more students complete 4 years of high school mathematics in preparation for college? As Adelman (2004) suggested, 4 years of such mathematics at the high school level is the best indicator of a student's likelihood of graduating from college.

As a community, it is important to continue to engage in research dedicated to a deep understanding of the nature and source of student differences in mathematics, and the identification of appropriate responses to the conditions thought to underlie those differences. The objective should be to document instructional materials and practices that promote the goal of more students learning more mathematics more effectively.

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